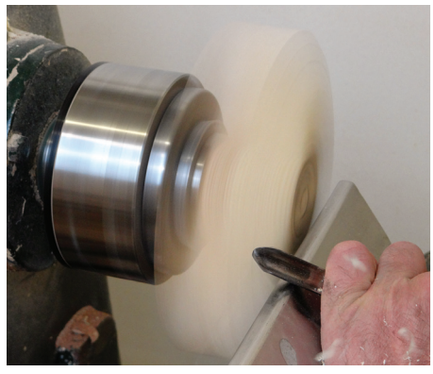
Understanding grain direction

No matter whether you are turning a large tabletop, laminated from narrower boards to form the large surface, or a tiny rosette or drinks coaster, the grain will be running across the lathe. This means that as it rotates, you will be turning side grain, followed by end grain, followed by side grain and then again by end grain, and so it goes on. This combination of easy-to-cut side grain combined with end grain, which is much harder to cut – as well as the difficult transition between the two, which is cutting in totally the wrong direction against the grain – is what makes this type of work so challenging. There is nothing you can do about this. It is a fact that twice in each rotation, you will be cutting the wood in completely the wrong direction, which goes against every principle of woodworking, but is simply a part of our lives as woodturners. Fortunately, there are a few things that we can do to lessen the harsh effects of this cutting against the grain action.

As you can see from the photographs, cutting square to the end grain is the least good option. This is why bowls made with a smooth flowing curve cut and finish easier than a square-sided bowl, often made by beginners. By working across the end grain in a curve, it reduces the harshness of the cut, as can be seen in the pictures. Sometimes a square edge can't be avoided, especially if a design calls for it. However, if you can, it will be easier to achieve a clean cut on a curved surface.

Faceplate work is usually circular, but mounting a piece of timber in this fashion really emphasises the problems of grain direction

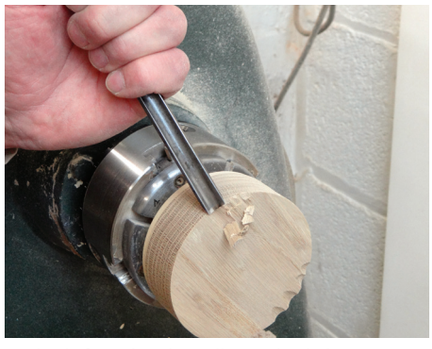
...leaves a very torn surface. You can see here exactly the point where the timber is worked against the grain and tears



By cutting in the correct direction and in a curve...

...you can see just how much more cleanly the wood cuts

Direction of cut



Cutting in from the edge splinters the timber...

It is important to understand the direction you should be cutting in. With spindle turning you will cut from the top of each shape, downhill, to the bottom. This is also correct for faceplate work, but because the grain is running across the lathe rather than with the lathe bed, the top of the 'hill' is not in the same place as it is for spindle work. The top of the hill is on the face of the wood, rather than the edge. If you are ever in doubt as to which way to cut, imagine you are going to carve the work with a carving gouge and mallet. If you start at the edge and cut in from there, you will splinter the wood, cutting against the grain. To cut with the grain you will need to work away from the edge.

...whereas cutting out from the centre produces clean chips and leaves a clean surface finish

The comparison leaves little doubt as to the direction in which you need to cut

MYTH BUSTERS: 'FACEPLATE WORK SHOULD ALWAYS BE DONE WITH A BOWL GOUGE'

Generally speaking, I will keep bowl gouges for bowls and spindle gouges for spindles, but faceplate work falls somewhere between the two, often featuring similar turned details to those found on spindles but on a piece of wood that closely resembles a bowl blank. A bowl gouge is a strong tool, designed to reach over the toolrest and hollow out bowls, but faceplate work rarely has to be hollowed like this, so the toolrest can usually be kept close to the work, which means that there is no need for the deep flute of a bowl gouge. The shallow fluted spindle gouge will often be easier to use to form these details. The choice is really yours for this kind of work; my preference is for a spindle gouge but if you are more comfortable using a bowl gouge, then that's fine too. Never use a spindle roughing gouge though, as it isn't designed for this kind of work. As long as you work safely and you can make the shapes that you want with the tool, then go for it.

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